

Exercise 7: Financial Forecasting

Objective

To implement a recursive algorithm that predicts the future value of an investment based on the present value, annual growth rate, and number of years.

Problem Statement

A financial forecasting tool predicts the future value of an investment using historical growth rates. The objective is to calculate the future value using recursion and analyse the algorithm's time complexity.

The task is to:

- Understand recursion.
- Create a recursive method to calculate future value.
- Display the predicted value.
- Analyse the time complexity.
- Suggest possible optimizations.

Theory

Recursion

Recursion is a programming technique in which a method calls itself to solve a problem. It breaks a large problem into smaller subproblems until a base condition is reached.

Advantages

- Easy to understand.
- Reduces code complexity.
- Suitable for mathematical and divide-and-conquer problems.

Disadvantages

- Uses additional memory because of function calls.
- Can become slow if many recursive calls are made.

Algorithm

1. Read the present value.

2. Read the annual growth rate.
3. Read the number of years.
4. If the number of years is 0, return the present value.
5. Otherwise, call the same method for one less year.
6. Multiply the returned value by **(1 + growth rate)**.
7. Display the future value.

Program

```
using System;

public class Program
{
    // Recursive function to calculate future value
    static double PredictFutureValue(double presentValue, double growthRate, int
years)
    {
        if (years == 0)
            return presentValue;

        return PredictFutureValue(presentValue, growthRate, years - 1) * (1 +
growthRate);
    }

    public static void Main()
    {
        Console.Write("Enter Present Value: ");
        double presentValue = Convert.ToDouble(Console.ReadLine());
        Console.Write("Enter Growth Rate (in %): ");
        double growthRate = Convert.ToDouble(Console.ReadLine()) / 100;
        Console.Write("Enter Number of Years: ");
        int years = Convert.ToInt32(Console.ReadLine());
```

```

    double futureValue = PredictFutureValue(presentValue, growthRate,
years);

    Console.WriteLine("\n----- Financial Forecast -----");
    Console.WriteLine("Present Value : " + presentValue);
    Console.WriteLine("Growth Rate : " + (growthRate * 100) + "%");
    Console.WriteLine("Years : " + years);
    Console.WriteLine("Future Value : " + futureValue);

    Console.WriteLine("\nTime Complexity : O(n)");
    Console.WriteLine("Space Complexity : O(n)");
}
}

```

Output:

```

Output
Enter Present Value: 10000
Enter Growth Rate (in %): 10
Enter Number of Years: 5

----- Financial Forecast -----
Present Value : 10000
Growth Rate : 10%
Years : 5
Future Value : 16105.1

Time Complexity : O(n)
Space Complexity : O(n)

=== Code Execution Successful ===

```

Conclusion

This exercise demonstrates how recursion can be used to calculate the future value of an investment. The recursive solution is simple and easy to understand, but it consumes additional stack memory. For larger values of years, an iterative approach or the mathematical formula is more efficient.